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Benefits and challenges of conversational agents in older adults

A scoping review

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In this article

Abstract

Background: Commercial conversational agents (CAs) bear the promise of low threshold accessibility for individuals with limited digital competencies. This applies not only for healthy aging older adults but also for specific subgroups such as those with life-long intellectual disabilities (ID).
Objective: This scoping review aims to synthesize the current evidence on benefits and challenges of CAs for older adults with and without ID. In doing so, we hope to inform future research as well as practical decision-making in the context of CAs as potential quality of life enhancers for older adults with various competence levels.
Material and methods: A literature search was conducted in form of a scoping review. A total of 841 publications were screened for benefits and challenges of CAs, resulting in an extraction of 18 articles targeting healthy aging older adults (60 years+) and 5 articles targeting older adults with ID (50 years+) for synthesis.
Results: The existing evidence suggests that CAs come with more benefits than challenges, e.g., general ease of use, easier information access, and feelings of companionship. Higher perceived agency due to using a CA seems to be a specific issue for older adults with ID. Challenges concern mostly learning how to use a CA and privacy concerns.
Conclusion: The results indicate that CAs can serve as quality of life enhancers both in healthy aging adults and in older adults with ID; nevertheless, thoughtful preparation is necessary, especially in relation to learning needs, capabilities present and privacy concerns.

Keywords
Digital voice assistants · Smart speaker · Sociotechnical systems · Old age · Digitalization

Supplementary Information

The online version of this article (https://doi.org/10.1007/s00391-022-02085-9) contains supplementary material, which is available to authorized users.

Introduction

Conversational agents (CAs), also known as voice assistants or speech assistants, "enable people to interact with smart devices using spoken language in a natural way" [11, p. 1] and they are potentially accessible for individuals with limited digital competencies, such as a large portion of older adults (e.g., [25]). In the last decade there has been an increasing spread of CAs in the commercial sector, including Amazon's Alexa (Amazon.com Inc., Seattle, WA, USA), Apple's Siri (Apple Inc., Cupertino, CA, USA) or Google Assistant (Google LLC, Mountain View, CA, USA), with sales rates amounting to 40 million units worldwide in the first quarter of 2022 [24]. In Germany, 14% of seniors used CAs in their daily life by the year 2021 [12]. However, CAs may also come with relevant challenges for older adults, e.g., when it comes to the installation of new devices, the customization of user preferences or when handling updates.

This paper reviews the available empirical research related to CAs with a focus on benefits and challenges for older adults, a societal group known to be at the risk of digital exclusion (e.g., 25% of US older adults did not use the internet in 2021 [15]). We decided to focus on "healthy aging" (see also [26]) individuals (60 years and older), hence, older adults with normative age-related loss, but not with very severe health losses and pronounced multimorbidity or frailty. Healthy aging individuals are among the early adopters of technological innovations within the group of older people [8]. Since the adoption rate of CAs among older people is still relatively low [12], we expect that most can be learned from this group in terms of benefits and challenges at the moment. The experience of actual CA use in the group of healthy aging older adults can serve as an exemplary analysis that can later be extended and contrasted to more frail groups of older adults. In particular, we excluded persons with dementia (PWD) for the following reasons. First, speech assistance may not be the best option for PWD as they typically have difficulties with language capabilities needed for successfully using a CA. Second, the focus of this special section on commercially available devices also implies that older adults themselves buy the devices based on their informed consumer decision. This might be a challenge for PWD; however, given the specific topic of our ongoing research and researcher expertise involved [21, 22], we also consider research targeting older adults with life-long intellectual disabilities (ID),[1] including older adults with ID 50 years and older. Older people with ID have received little attention in research on CAs so far (e.g., [21, 23]) and they face a particular risk of digital exclusion [8], CAs, however, may come with pronounced advantages for this specific subgroup of older adults, as they may offer support in activities of daily living for older adults with ID, enhance their ability for independent living and increase their social participation [21, 23]. On the other hand, older adults with ID may also face specific challenges in the interaction with CAs due to communicative restrictions. In this way, emerging evidence for this group might serve as a contrasting example to healthy aging individuals and eventually also inform research on CAs with older adults in general.

In contrast to two recent reviews on CA use in older adults taking a more general perspective [2, 9], we decided to focus our analyses on benefits and challenges, as this might especially inform ongoing research and expertise debates about promoting digital inclusion of older adults or serve for informing decisions of practitioners and stakeholders when it comes to planning the deployment of commercial CAs in the lives of older adults with and without ID.

Research questions

We address two primary research questions:

- RQ1: What are the reported benefits and challenges of CAs for healthy aging older adults?
- RQ2: What are the reported benefits and challenges of CAs for the specific and previously largely neglected subgroup of older adults with ID?

Methods

We conducted a systematic literature search with the end of May 2021 as deadline in the following databases: ACM Digital Library, IEEE Xplore, ERIC, PubMed, PsycINFO, and Web of Science. The search was limited to peer-reviewed articles published in English since 2010, as the emergence and spread of commercial CAs only happened after 2010. A minor number of articles were found additionally based on a manual reference research and Google Scholar was used to find additional potentially relevant studies.

Search terms

Appendix 1 shows the applied search string. The search strategy consisted of a combination of synonyms for the two main search areas "conversational agents" and "older adults" with an extension for the special group of interest, i.e., older adults with ID.

Study selection

We followed the guidelines for systematic scoping reviews suggested by Peters et al. [14]. **Figure 1** shows a flow diagram of the screening process. Titles and abstracts were screened for eligibility by CE and the selection was verified by TH and AS. Full-text screening followed the same approach. In cases of different opinions about inclusion and exclusion, studies were discussed in the review group to reach a consensus.

Inclusion and exclusion criteria

We included empirical studies focusing on older adults' perspectives, requirements, and user experiences of commercial and prototype CAs. The age threshold for healthy aging older adults was set at 60 years (i.e., minimum age of participants or mean age of sample being ≥ 60 years). For studies focusing on individuals with ID, we applied an age threshold of 50 years (i.e., at least one participant with ID being ≥ 50 years) based on the fact that people with ID experience an earlier beginning of aging processes (e.g., higher levels of health needs). In addition, challenges of living as independently as possible arise in those with ID already in midlife due to their specific work and living situation.

We excluded CAs in form of telephone-based interactive voice response systems as well as robots, as both technologies differ strongly in the range and kind of available functions due to their technical setup (telephone) or physical embodiment (robots). Moreover, robots have a more distinct social presence than CAs. Virtual agents that had no option for speech input were also excluded. Concerning the study design, we excluded studies that analyzed customer reviews or role plays or focused primarily on technical aspects

[1] "Intellectual disability is a condition characterized by significant limitations in both intellectual functioning and adaptive behavior that originates before the age of 22 years." [19].

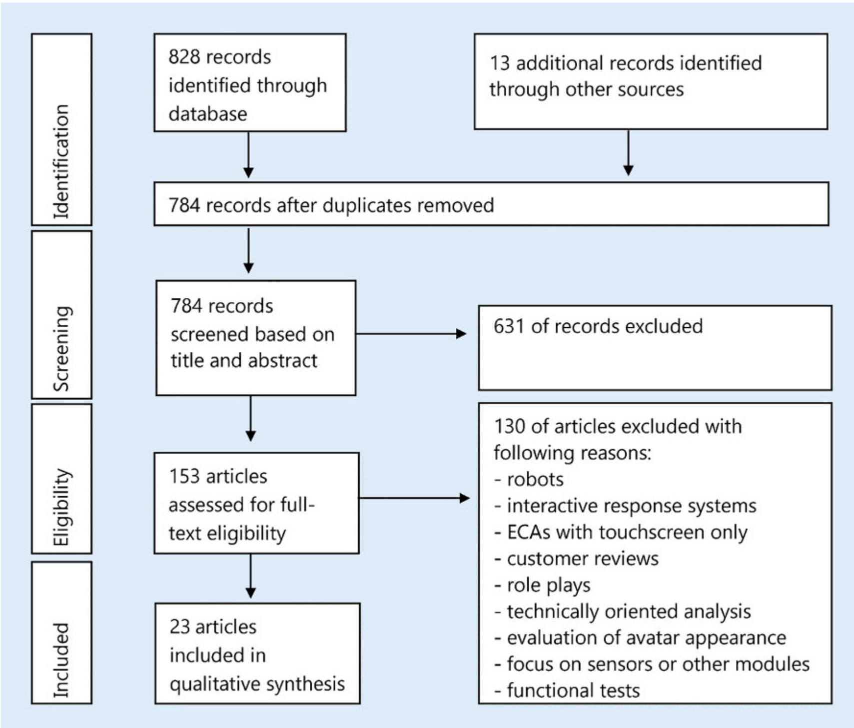

Fig. 1 ▲ Flow diagram for study selection. *ECAs* Embodied Conversational Agents

concerning (dialog) design, avatar appearances, or specific system modules (see Appendix 2).

Results

The screening process resulted in 23 studies for synthesis. There were no studies that investigated both envisaged user groups at the same time. Therefore, results are presented separately for both groups. In total, 18 publications focused on healthy aging older adults, while 5 studies focused on older adults with ID. Appendix 3 gives an overview of the included publications with assigned publication identifier (A01–A23). Studies investigating older adults with ID are marked with an asterisk (e.g., A01*).

Study characteristics

Key characteristics of the included studies are presented in Appendix 4.

Study population. Of the 18 studies targeting healthy aging older adults 11 reported no specific health conditions among the participants, while 7 studies included older adults with circumscribed age-related impairments or diseases like hypertension, diabetes, visual or other physical impairments, and chronic lung diseases (A03, A04, A09, A11, A16, A18, A19). The mean age of participants across all publications was 72.8 years (age range: 50–97 years).

Although the information that participants in 5 publications were characterized by ID was clear, the given descriptions of the form and grade of ID were very diverse. Of the papers 4 provided specific information on the nature of the ID (A02*, A08*, A10*, A17*) while 1 article did not provide such information (A01*). The mean age of participants in the ID subsample was 43.7 years (age range: 18–79 years).

The majority of studies with healthy aging adults employed commercial CAs ($n = 12$), including Amazon Alexa, Apple Siri and Google Assistant, Clova (Naver Corporation, Bundang-gu, Seongnam, South Korea), and NUGU (SK Telecom, Seoul, South Korea).

All 5 studies focusing on older adults with ID deployed commercial CAs.

Study environment and duration of usage. A total of 21 studies used either a commercial CA ($n = 17$) or prototype CA ($n = 4$), while 2 studies reported only of requirements for a to be designed CA. Of the 21 user studies, 6 took place in private households, 9 in laboratories. The remaining 6 studies took place in rooms of the residence or facility in which the participants lived or were associated with.

Of the 21 user studies 9 presented user experiences of participants who used a CA more than once. Duration of usage covered very different time periods, ranging from 1 week to 1 year.

Data collection methods. The majority of studies (12 of 23) applied a qualitative approach, using structured observations, audio or video recordings of CA use, semi-structured interviews or focus group interviews, 7 studies used a mixed methods approach (A03, A10*, A12, A14, A15, A17*, A21) and 4 studies applied quantitative analyses, including structured interviews and/or questionnaires and test scores (A02*, A05, A08*, A22).

Findings on benefits and challenges related to CAs use

A 3-layer scheme for in-depth analysis of reported benefits and challenges of CAs for older adults. In an inductive approach, we developed a coding structure based on a differentiation of three user experience levels. Under the first level, i.e., the level of speech exchanges with the CA, we summarized results that refer to the evaluation of device-related features or the general evaluation of voice control. The second level, i.e., the integration of the CA into everyday life, focuses on the range of CA functions that users embedded into their everyday life routines as well as reflection on (un)successful usage or integration. At a third level, i.e., dimensions of quality of life affected by the conversational agent, we compiled results that pertain to a user's perception or affective state, e.g., feelings of companionship or social integration or worries when using a CA.

Range of benefits and challenges of CAs in older adults with and without ID

◘ **Table 1** presents the results. Appendices 5 and 7 show the same table ex-

Table 1 Perspectives and experiences of older adults with and without ID in CA use

Analytical categories		*n* (OA)	*n* (OA with ID)
(1) Level of speech exchanges with the CA			
Benefits	***Good/high usability:***	**13**	**2**
	Easy	13	2
	Hands-free interaction	6	1
	No typing necessary	5	1
	Fast	4	1
	No worries about spelling	2	1
	Friendly communication style	**3**	**–**
Challenges	***Problems with usability:***	**9**	**3**
	Wording difficulties	4	3
	Device-incompatible timing	3	2
	Speech recognition problems	4	–
	Time consuming	2	–
	General rejection of speech control mode	**2**	**–**
	General pronunciation problems	**–**	**4**
(2) Integration of the CA into everyday life			
Benefits	***Internet search and information access***	**8**	**1**
	Time management	**5**	**1**
	Health management	**7**	**–**
	Entertainment and leisure time	**5**	**2**
	Communication with:	**6**	**2**
	The conversational agent	6	1
	Other people	2	1
	Smart home features	**3**	**1**
Challenges	***Learning to use the CA***	**4**	**1**
	Compatibility with other tools	**3**	**–**
	Breaking established habits	**2**	**–**
	Getting specific health-related information	**1**	**–**
(3) Dimensions of quality of life affected by the CA			
Benefits	***Feeling of companionship from CA***	**6**	**1**
	Higher perceived security	**4**	**–**
	Higher perceived independence	**2**	**1**
	Lowered feeling of loneliness due to communication with people	**1**	**–**
	Higher perceived agency	**–**	**1**
Challenges	***Source for different kinds of worries/ concerns:***	**10**	**1**
	Privacy concerns	6	–
	Fear of losing competencies/capabilities	3	–
	Security concerns	2	–
	Worry about agent-related costs	2	–
	Being afraid of the agent itself	1	–
	Being afraid of crashing the program	–	1
	Lack of trust in the CA's answers	**4**	**–**
	Lack of perceived usefulness	**3**	**–**

OA older adults, *ID* intellectual disabilities, *CA* conversational agent

panded by a list of publications of which the results were extracted. Appendices 6 and 8 present the range of benefits and challenges in relation to market readiness of the CA and duration of usage by the study participants.

Level of speech exchanges with the CA

Benefits. In total, 13 of the 18 articles focusing on healthy aging adults reported on benefits related to aspects of good usability. Most frequently, the voice-based interaction was perceived as generally easy or easier compared to other digital devices. Regarding the group of older adults with ID, 2 of the 5 papers presented results related to good usability.

Challenges. The most often reported challenges for healthy aging older adults also concerned usability aspects including wording difficulties when using commercial CAs (e.g., forgetting the wake word or specific keywords or commands, producing too long or complex utterances, device-incompatible timing, and speech recognition problems). Challenges for older adults with ID were also wording difficulties ($n=3$) and timing problems ($n=2$). A specific barrier only reported for older adults with ID were general pronunciation problems ($n=4$). However, it was also reported that these could be overcome partly by training (A01*, A17*).

Integration of the CA into everyday life

Benefits. A total of six subcategories were extracted for healthy aging individuals. Using the CA for Internet search and information access was the most frequently identified subcategory ($n=8$). All benefits identified for healthy aging older adults except for health management were also reported for older adults with ID.

Challenges. The most relevant challenges for healthy aging adults were difficulties in learning to use the CA ($n=4$) and a lacking compatibility with other devices ($n=3$). One study focusing on older adults with ID (A17*) also reported learning difficulties, especially with respect to the functional scope of the device.

Dimensions of quality of life affected by the CA

Benefits. The most frequently identified aspect for healthy aging adults was a feeling of companionship from using the CA ($n=6$). One paper (A17*) presented benefits for older adults with ID in this domain with relation to feelings of companionship and higher perception of independence. A specific benefit for older adults with ID was the finding that the CA empowered people with ID to engage in activities without the otherwise necessary assistance by professionals (higher agency).

Challenges. In total, 10 papers reported that the CA was a source for various kinds of worries and concerns in the group of healthy aging adults, mostly related to privacy concerns ($n=6$). Additionally, 4 papers mentioned the lack of trust in the CA's answers as a barrier. For the group of older adults with ID, one study was identified in this domain, mentioning the worry of crashing the program (A10*).

Discussion

The goal of this scoping review was to give an overview of benefits and challenges of CAs for healthy aging older adults, enriched with a focus on older adults with ID. Out of 841 papers identified by a systematic literature search, we included 23 articles into a synthesis, of which 17 focused on commercial CAs. We proposed a categorization scheme on three levels of experiences with a CA to synthesize findings.

Overall, the available evidence suggests more benefits than challenges in CA use for older adults. Additionally, many of the identified benefits and challenges are comparable for older adults with and without ID. Specific aspects for older adults with ID concern the empowerment to engage in activities without assistance on the level of benefits and more general pronunciation problems on the level of challenges.

Concerning benefits on the level of speech exchanges with the CA, our findings support the benefit of natural and easy communication with CAs suggested by the industry, and by studies which have shown that voice-based interaction implies a higher accessibility for persons with sensorimotor impairments [1, 16, 17]. On the other hand, the reviewed literature also suggests that such a positive evaluation might be too general in the light of the reported usability problems in both groups of older adults, and especially for commercial CAs. Learning difficulties and a need for training about how to use a commercial CA were identified [23]. However, the actual learning process and the training needs had not been the focus of the reviewed studies and the question remains unanswered what kind of communicative problems arise, e.g., with respect to word choice, grammar and speech acts. This is in contrast to research on older adults' learning experiences when using other digital devices, which have been investigated quite extensively [4, 10, 13, 20]. Future research should focus on the detailed analysis of the learning processes of older adults in the context of commercial CAs and how to overcome the identified usability challenges. This desideratum is also consistent with the conclusion of the recent systematic review by Jakob [9] on the "gap in research in the context of the independent learnability of VCDs and older adults" (voice controlled devices; [9, p. 189]).

A starting point to support the learning process could be to provide a basic tutorial on the functioning of a CA with a focus on the programmed language and action model of the device which is always more restricted than the human competence of interpreting actions. This might sensitize users to the restrictions of CAs in terms of language processing and management of reactions to user input. This learning topic might then be complemented by a tutorial on strategies for overcoming problems in speech exchanges with a CA. Such a training concept could be generated based on the vast conversation analytical body of research on repair in social interaction [6].

On another practical level, this review might serve as a first orientation for practitioners and stakeholders when it comes to planning the deployment of commercial CAs in the lives of older adults with and without ID. An important recommendation would be to prepare thoughtfully for training, support and learning materials, adequately designed, e.g., in large print, easy to read language and clear illustrations. To some extent, such learning material is available on websites of nonprofit associations (e.g., the Federal Working Group for Older Adults [3] or The Digital Angel [7]).

Another important challenge in this context might be the identified privacy concerns, which are a relevant barrier to using CAs among older adults, especially in the context of commercial CAs. These aspects have already started to gain attention in current research projects. Application-oriented approaches aim to develop technical solutions to strengthen users' control over their data. As an example, a webpage-based tool for improving data sovereignty and empowerment is already available [5] and the development of a meta-assistant for data autonomy in CA use is being worked on [18]. Using such additional tools might be a reasonable and practical way to raise awareness for these aspects and to handle privacy concerns.

Limitations

Although we applied an extensive search strategy, we cannot completely rule out the possibility that other relevant studies are missing, e.g., studies in languages other than English or German, grey literature, and studies not listed in the searched databases. Although English has been established as lingua franca for scientific publications worldwide, there might be a bias towards research in Western, European and Western-oriented countries. Furthermore, analyzing cultural and linguistic features in different countries or for different languages is an important topic that was beyond the scope of this review and should be considered in future work. Also, our search deadline was May 2021. Relevant research that might have appeared in the meantime may possibly lead to additional differentiations.

Conclusion

CAs have the potential to offer benefits for older adults with and without ID on different levels. Among the major challenges are the need for training, particularly speech-related instruction, how to manipulate a commercial CA such that user needs are fulfilled in most optimal ways, and privacy aspects.

Practical recommendations

- We recommend to apply the three levels differentiated in this review, i.e., level of speech exchanges with the CA, integration of the CA into everyday life, and dimensions of quality of life affected by the CA, as guidelines able to inform about practical implementation.
- Benefits of using CAs seem mostly clear for healthy aging older adults, but training and instruction including the thorough consideration of privacy concerns are necessary. Specific subgroups such as older adults with ID have an even higher need of training and instruction, which should be developed considering the specific demands and abilities within this group.

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Zusammenfassung

Nutzen und Herausforderungen konversationeller Agenten für ältere Menschen. Ein Scoping-Review

Hintergrund: Kommerzielle konversationelle Agenten (KAs) versprechen eine niedrigschwellige Zugänglichkeit für Personen mit eingeschränkten digitalen Kompetenzen. Dies gilt für gesund alternde ältere Menschen, aber auch für bestimmte Untergruppen wie Ältere mit lebenslanger geistiger Behinderung (gB).
Ziel: Dieses Scoping-Review zielt auf eine Synthese der aktuellen Forschung zu Nutzen und Herausforderungen von KAs für ältere Menschen mit und ohne gB ab. Damit soll eine Grundlage für Forschung und Praxis im Zusammenhang mit KAs als potenziellen Verstärkungsfaktoren für die Lebensqualität älterer Menschen mit verschiedenen Kompetenzniveaus geschaffen werden.
Material und Methode: Die Literaturrecherche wurde in Form eines Scoping-Reviews durchgeführt. Es wurden insgesamt 841 Publikationen gesichtet und 18 Artikel zu gesund alternden älteren Menschen (≥ 60 Jahre) sowie 5 Artikel zu älteren Menschen mit gB (≥ 50 Jahre) in die Synthese eingeschlossen.
Ergebnisse: Die Ergebnisse deuten darauf hin, dass KAs mehr Vorteile als Herausforderungen mit sich bringen, z. B. insgesamt eine einfache Nutzung, ein leichterer Zugang zu Informationen sowie das Gefühl von Gesellschaft. Eine höhere Wahrnehmung von Handlungsmacht aufgrund der KA-Nutzung scheint ein spezielles Ergebnis für ältere Menschen mit gB zu sein. Herausforderungen betreffen das Erlernen der Nutzung eines KA und Bedenken hinsichtlich der Privatsphäre.
Schlussfolgerung: KAs können die Lebensqualität sowohl bei gesund alternden Erwachsenen als auch bei älteren Erwachsenen mit gB verbessern. Dennoch ist eine sorgfältige Vorbereitung erforderlich, insbesondere in Bezug auf die Lernbedarfe, vorhandene Fähigkeiten sowie Bedenken zur Privatsphäre.

Schlüsselwörter
Digitale Sprachassistenten · Smart Speaker · Soziotechnische Systeme · Hohes Alter · Digitalisierung

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Declarations

Conflict of interest. C. Even, T. Hammann, V. Heyl, C. Rietz, H.-W. Wahl, P. Zentel and A. Schlomann declare that they have no competing interests.

For this article no studies with human participants or animals were performed by any of the authors. All studies mentioned were in accordance with the ethical standards indicated in each case.

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